CSA0412-OPERATING SYSTEMS

EXP 07-Design a CPU scheduling program using FCFS technique

```
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Enter burst time for P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];
```

```
    }

    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t\t%d\t\t%d\n",
               i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

    return 0;
}
```

```
Enter the number of processes: 3
Enter burst time for P1: 5
Enter burst time for P2: 3
Enter burst time for P3: 2

Process Burst Time      Waiting Time    Turnaround Time
P1      5               0               5
P2      3               5               8
P3      2               8               10

Average Waiting Time = 4.33
Average Turnaround Time = 7.67


...Program finished with exit code 0
Press ENTER to exit console.
```